$f(x,y,z)$ $(x, y, z \geq 0)$

Domain = $Z > 0$

Domain = half space of Z

Range = $(-\infty, +\infty)$

[Surface level set of points where function is constant

level curve → 2D
level surface → 3D

Theory OF Automata:

[Computer's Theory (Daniel I.A. Cohen)

Q: How first time C++ code were prepared for execution?

Q: How the Java code is prepared for execution on its host / host machine?

(Byte ~~class~~ code)

Java
↓
Java compiler
↓
.class (Bytecode)
↓
interpreter
↓
hardware

مشین کوڈ execute ہوتا ہے۔

(Interpreter) کے ذریعے execute کوڈ Byte

(1) when Java introduced?
(2) what is .netframework?

gap in market
microsoft.

Necessity is the mother of invention.

Java Speciality:
↳ write one code and run anywhere.

(A) Domain:
Application softwares
System Softwares

(3) machine language?
(4) is machine object oriented language?
(8) Object oriented?
Functional?
Procedural?

(B) what will be the paradyme of this language.
(C) what will be code process of

Naam

subject of TOA ✓
(Theory of automata)

Automata

plural of Greek word "automaton"

Automaton

Automaton means "machine" in english

So

Theory of machines {computer machines}

The most generic device that humans invented ever is compiler (created)

Software Engineering:

Software development life Cycle (SDLC)

Complex software System.

System Software

Application software

- mobile App
- web App
  - Brauser internet
- desktop App

Theory of Automata:

Language:

A total set of instructions given to a device is called as its language.

simple
- bulb
  - 1 On
  - 2 Off

Complex
- Fan
  - 1 ON
  - 2 OFF
  - ~~3 Slow~~
  - 3 Speed
    - slow
    - medium
    - Fast

Programming language:

Program
Collection of key strokes (characters)

Characters
↓
Symbol
↓
level
↓
Alphabet

A program is more a string

A collection of strings is called Programming langu

Programming language is ~~infinite~~ collection of different strings.

A program will be part of C++ language if it gets compile error free by a C++ compiler.

Q1 How a compiler comes to know that an inputed program is one from infinite valid programs belonging to that particular language?

Q2: Identifier:

(A user defined name) of any programming language construct.

1) letter
2) underscore
3) digit (0-9)
4) 2abex invalid.
4) It may start with underscore, letter only. after it any number of letters, digits can be appear in any order.

abc2 ✓ valid

جرمن، java، cpp، پنجابی، english، اردو

C#، Python

Formal languages: Java, Cpp, Python, C#

natural language: اردو, english, پنجابی, جرمنی

C# → microsoft, Java → sunmicrosystem

Easy Python, C#

(1,2,3) (First two chapters)

Theory of Automata

Language:

language is a collection of rules restrictions

Character Set:- (ASCII character set)

machine language character set → Binary

Every language comes with a character set

Sigma Σ

Sigma = All identifiers.

$\Sigma = \{a, b, c \ldots z, A, B \ldots Z, 0, 1 \ldots 9, _\}$

$\Sigma = \{0, 1\}$

$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Restrictions over using Sigma:

Rule 1: ① It must be non-empty set

② It must be finite

③ ~~Ø~~ Λ (capital lamda)

ε (Epsilon)

are forbidden to choose in sigma.

Length of a String:

~~AAA~~ (~~2~~) (~~1~~) aaa (1)

inherited → unavoidable:

This is the inherited property of all natural lange which is called ambiguity:

(زبان کی طبعی)

→ The count of positions (locations) available to hold symbols (pure theory)

↳ $\Sigma = \{a, aa, aaa, aaaa\}$

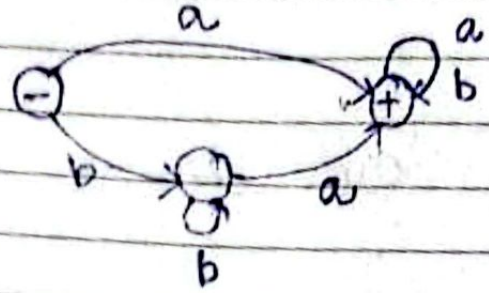
symbol symbol symbol

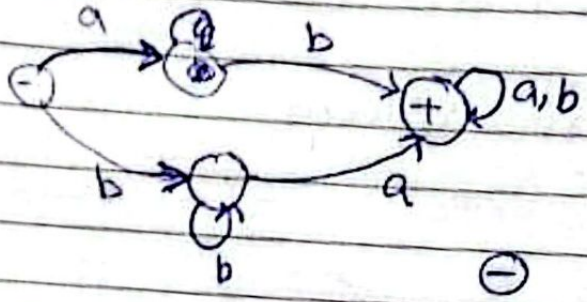
$\Sigma = \{2, 22, 222, 2222\}$

aaa → it may be 3 symbols or just one or two.

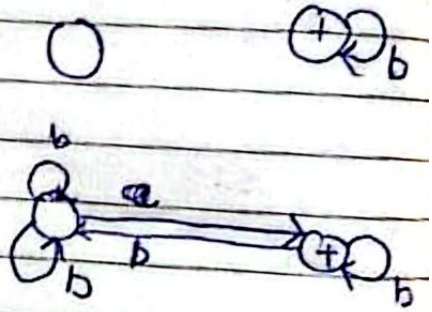
All the strings starting and ending with different letters;

$\Sigma = \{a, b\}$





$\Sigma = \{p, q$



$\Sigma = \{a, b, c\}$ H.W

All the string starting & ending with same letter.

$\Sigma = \{a, b, c\}$, $\Sigma = \{p, q, r\}$

Length:

All the strings whose length are divisible by



Character Count:

L: Count of a is even



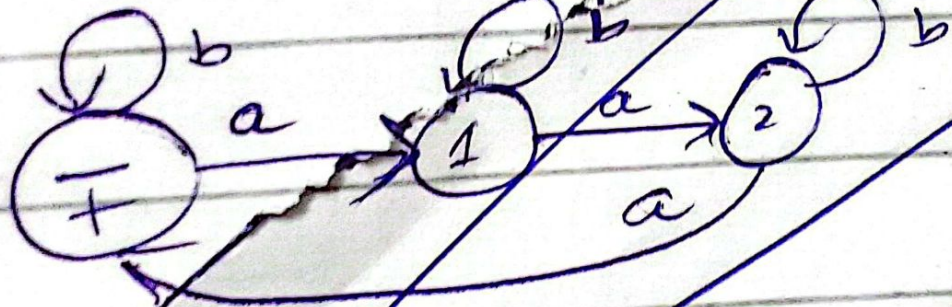
L: Count of b is odd:



Automata H.W

$\Sigma = \{a, b\}$

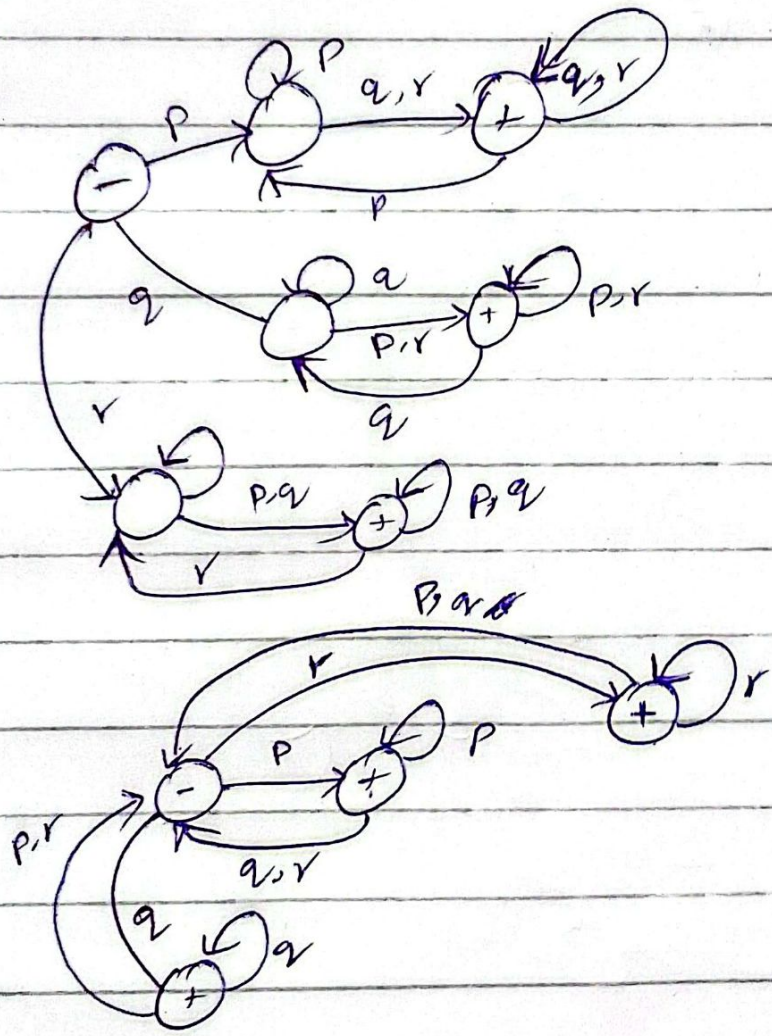
all strings where count of a is multiple of 3

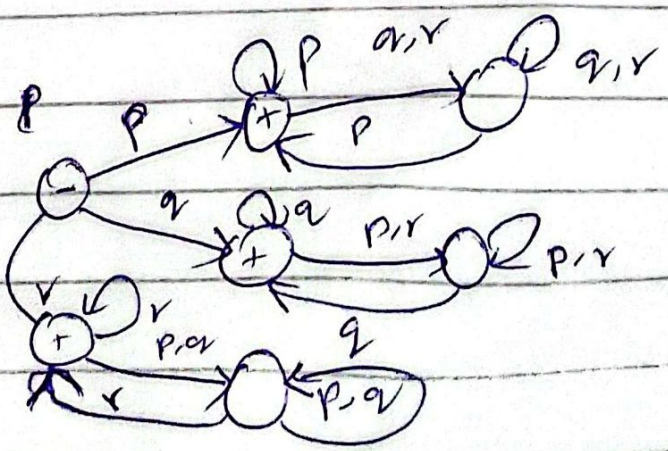


2nd last letter b

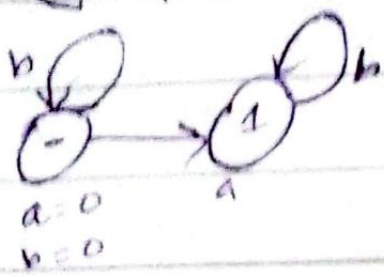
$\Sigma = \{a, b\}$

$\Sigma = \{p, q, r\}$ all strings starting and ending with diff

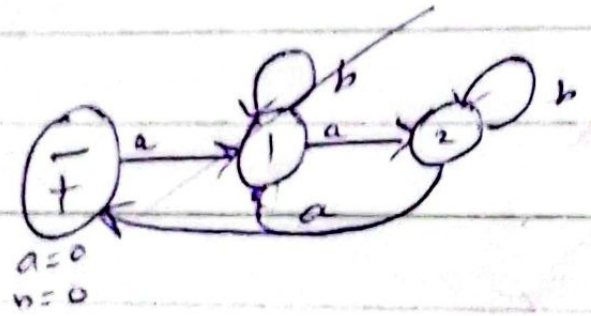


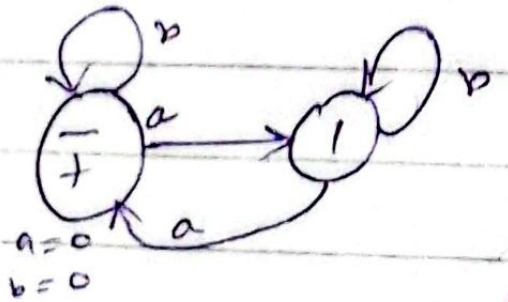


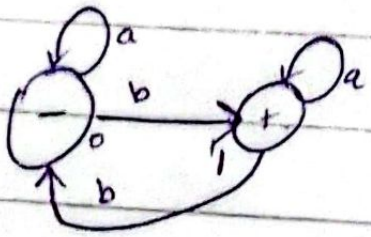
Character count

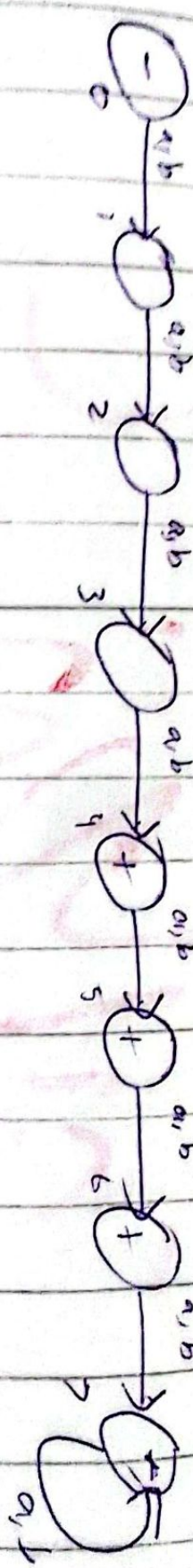


count of a = even



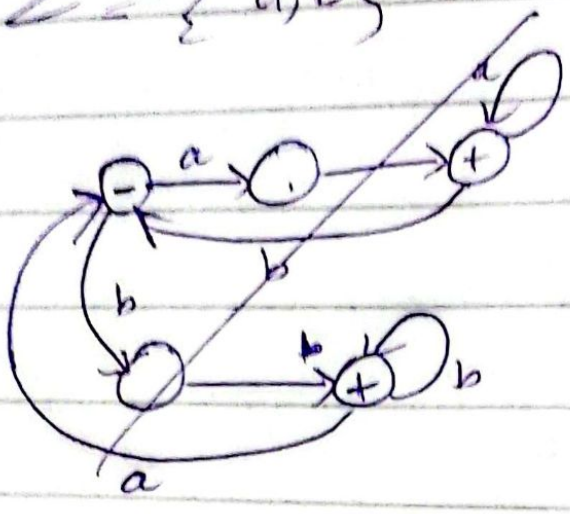






having double letters

$\Sigma = \{a, b\}$



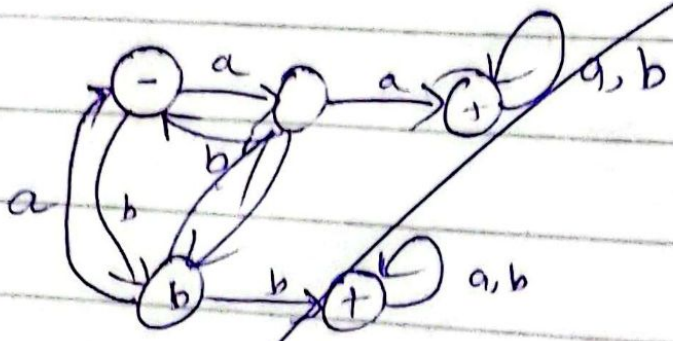
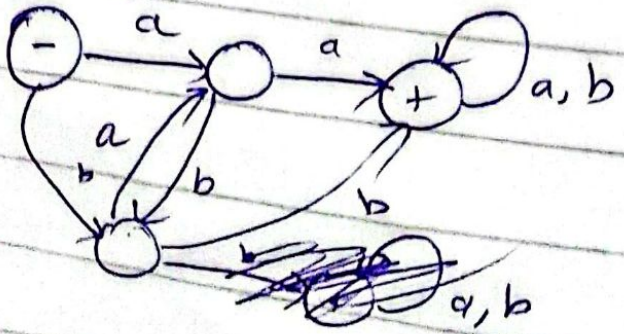
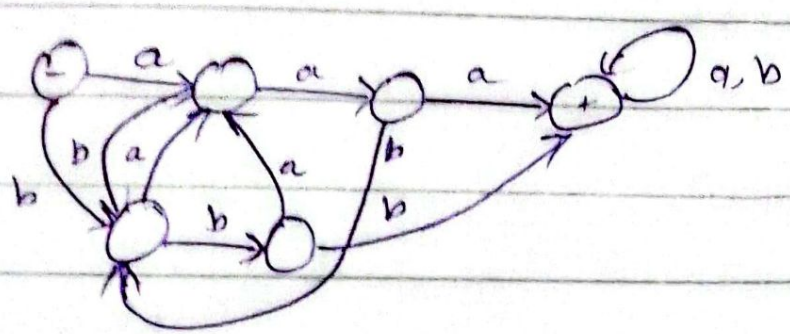
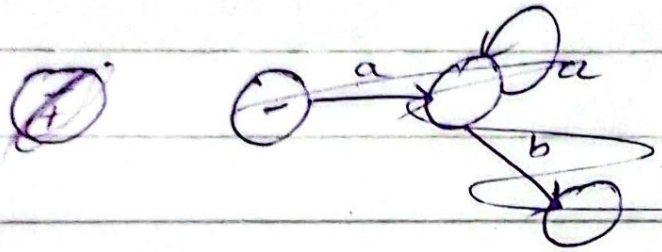


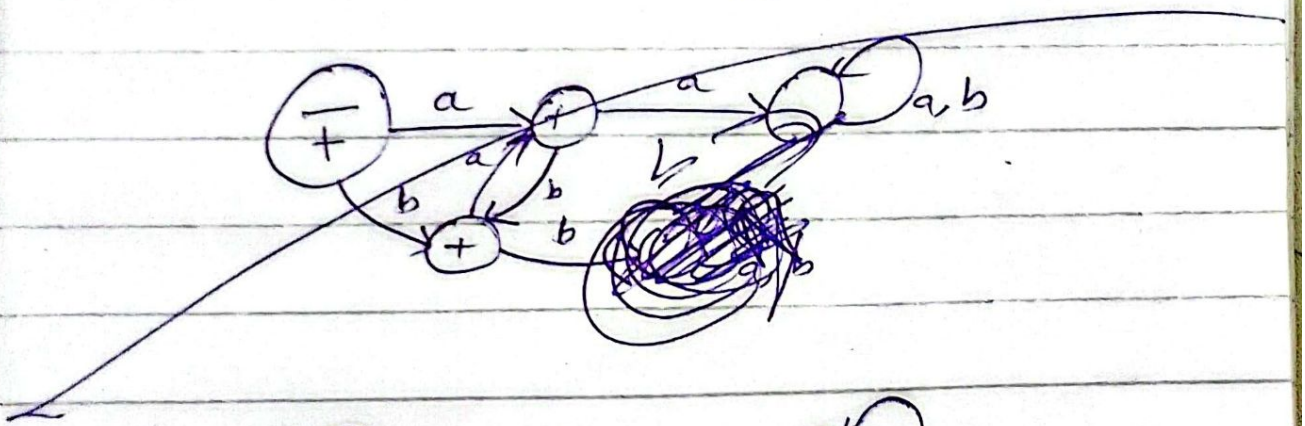
abb ✓

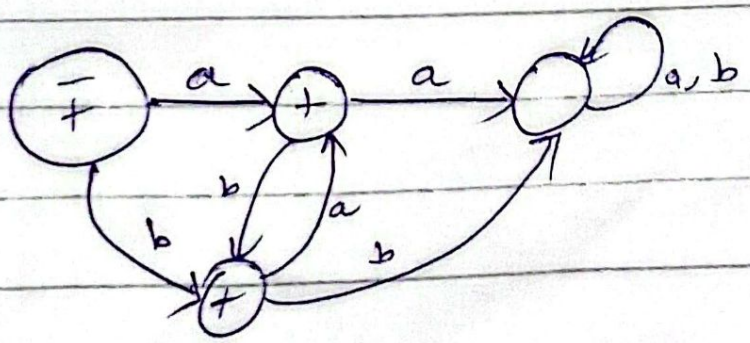


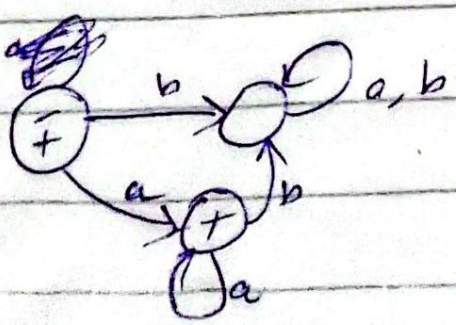
all strings having triple a











Automata

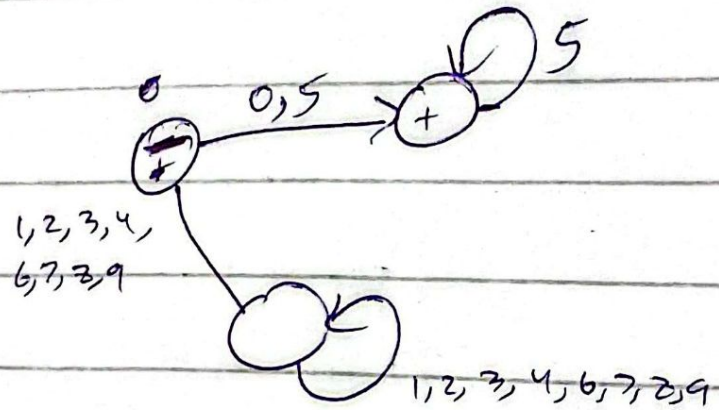
Klean's closure closure properties

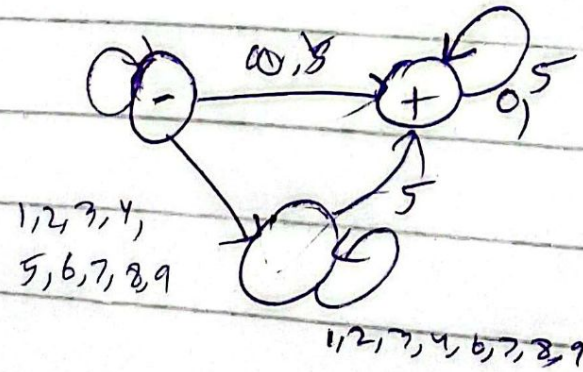
+ve closure = short hand notation.

$$a \cdot a^* = a^+$$

$$\Sigma = \{a, b\}^*$$

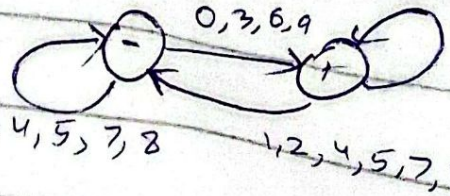
$$\Sigma^* = \{a, b\} = ?$$

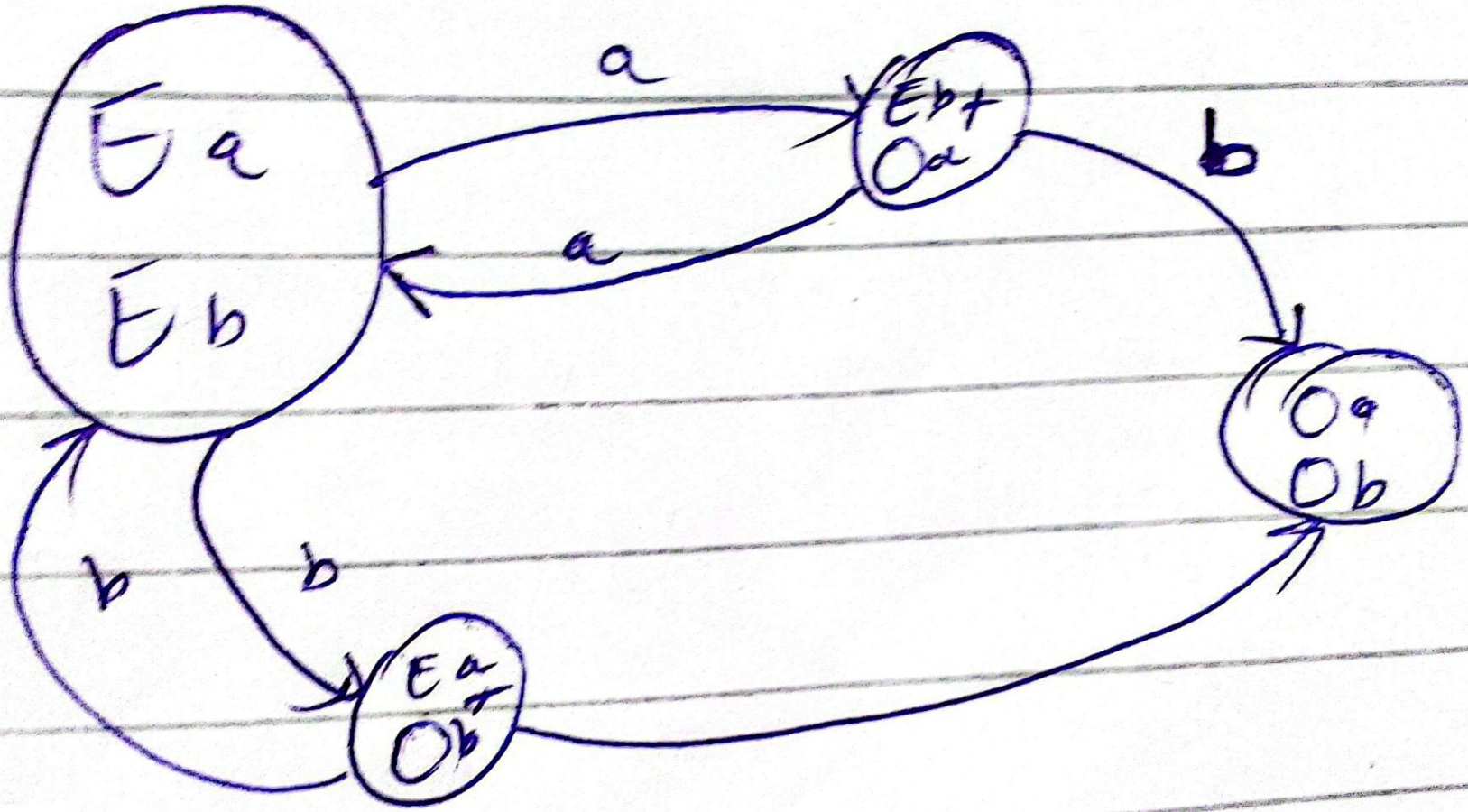


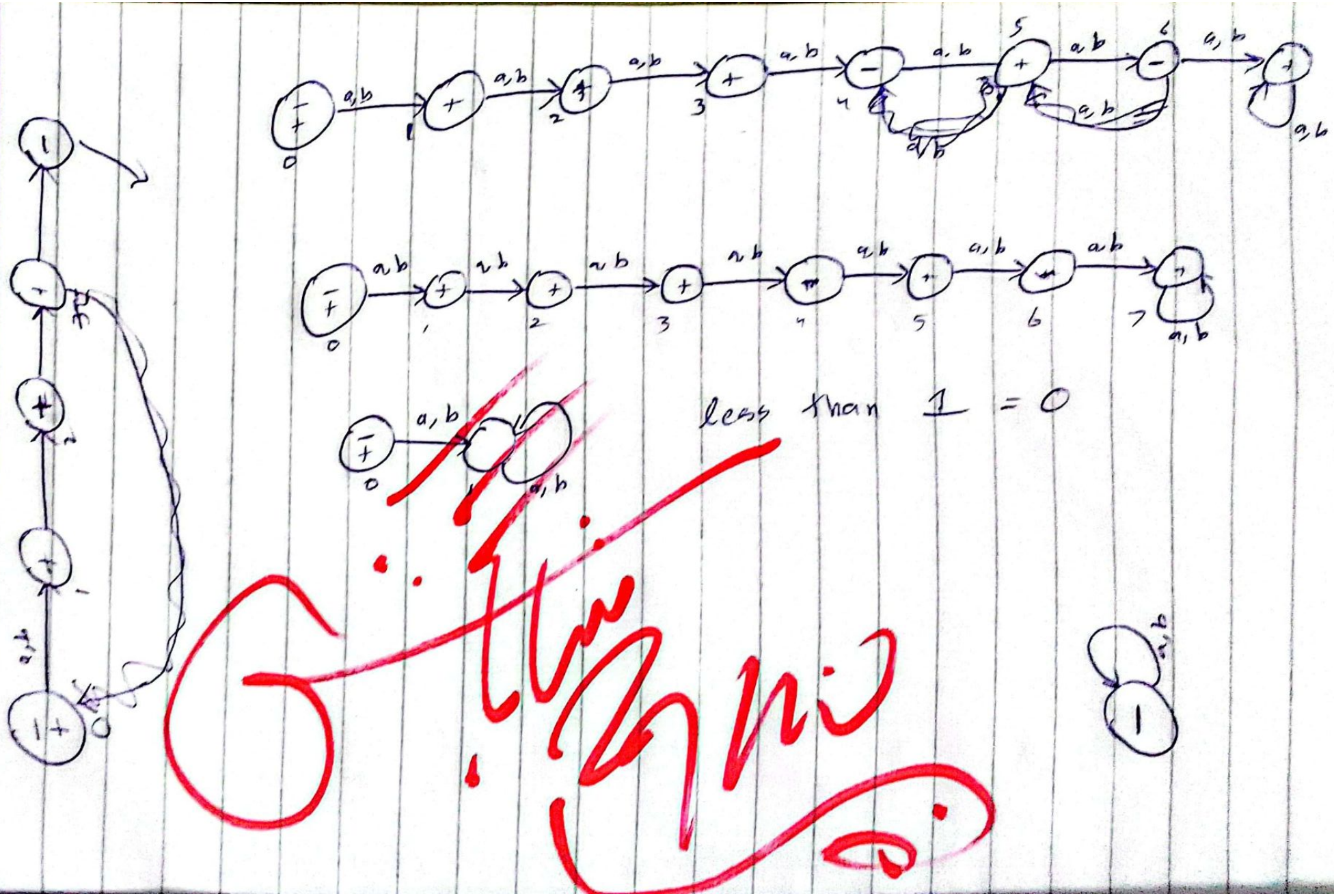


1,2,3,4

Divisble by 3



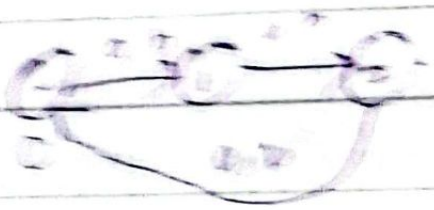

Ea
Eb
a
Eb+
Oa
a
b
Oa
Ob
b
b
Ea
Ob

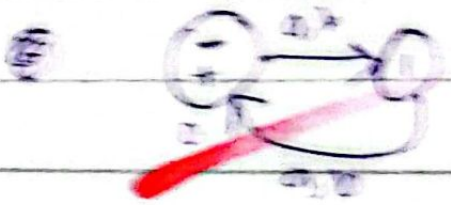

a,b
less than 1 = 0

Automata

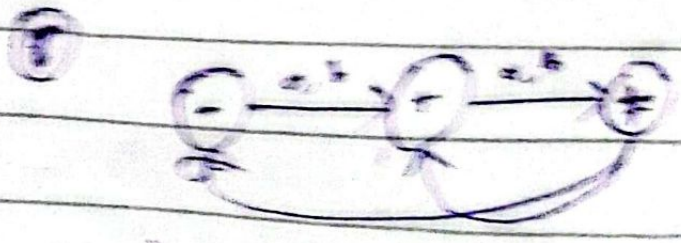
Length restriction













ba

1st & 2nd last letter must be different

2nd last is b for minimum

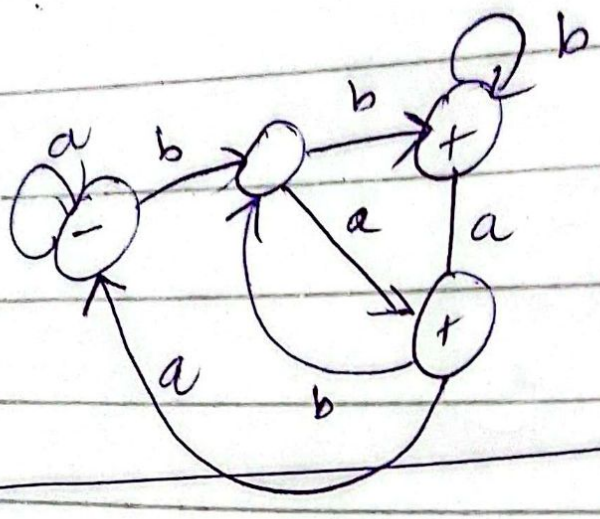
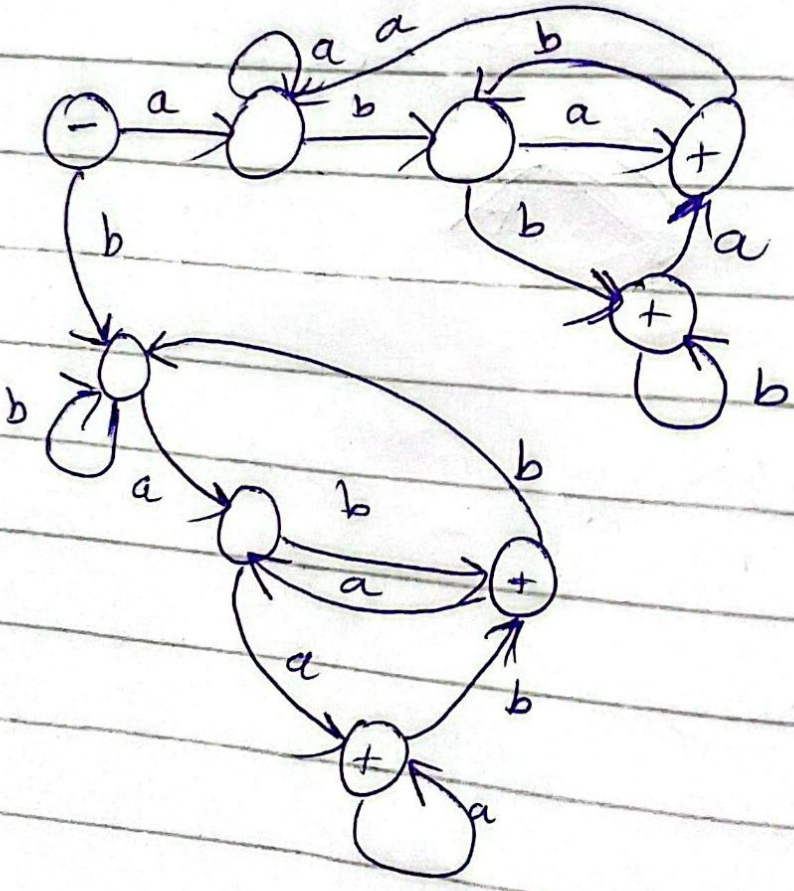


abb bbaab



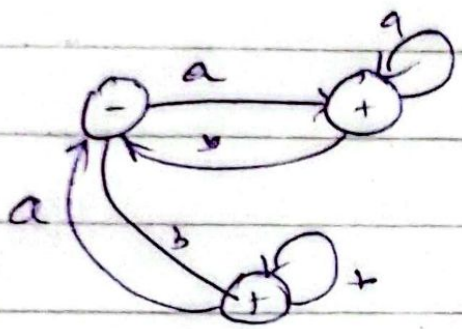
1st & 2nd last be diff.

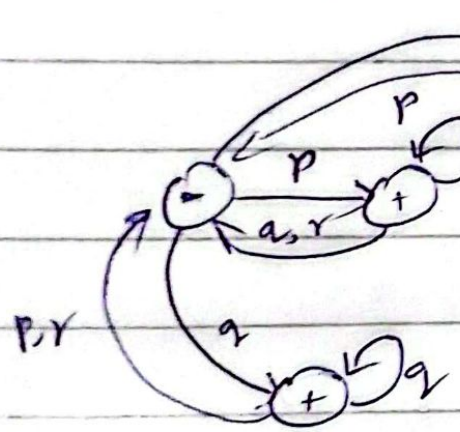
H.W

all strings starting & ending with some letters

$\Sigma = \{a, b\}$, $\Sigma = \{p, q, r\}$

F.A





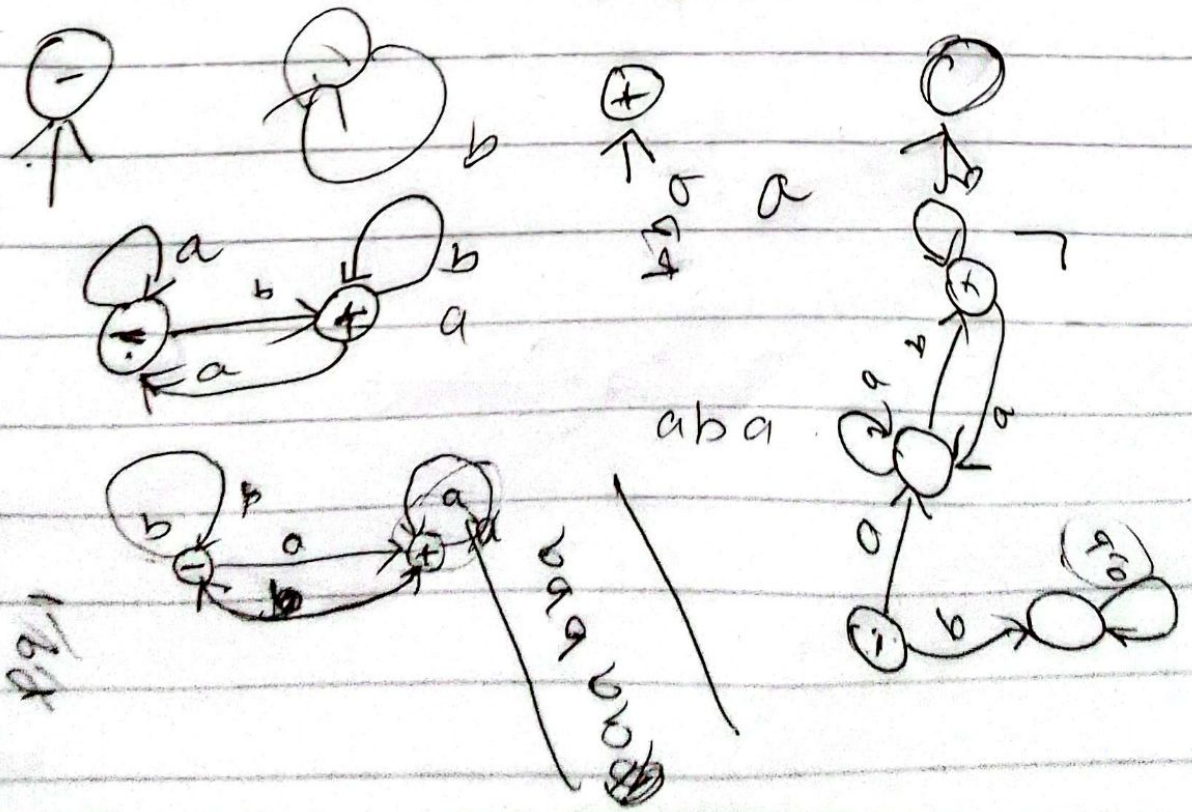
a,aa

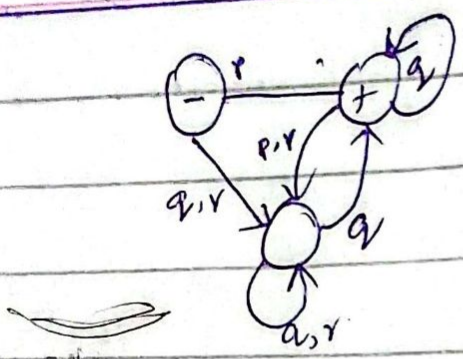
aab (ab) b

aabb

$\Sigma = \{a, b\}$

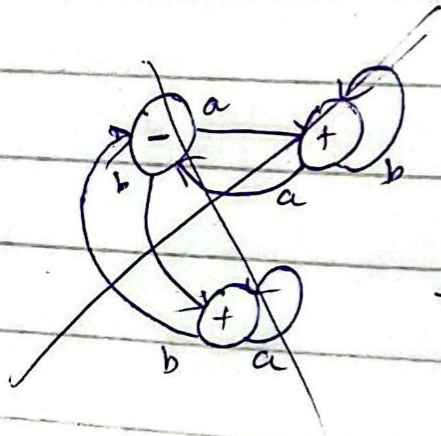
a



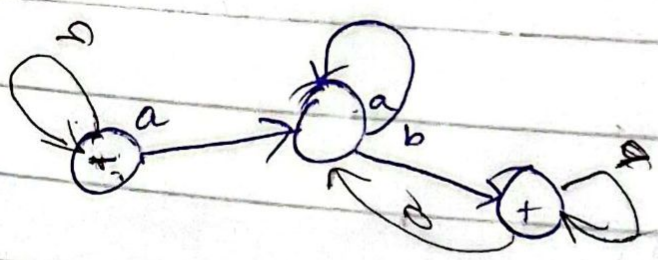


$\Sigma = \{a, b\}$

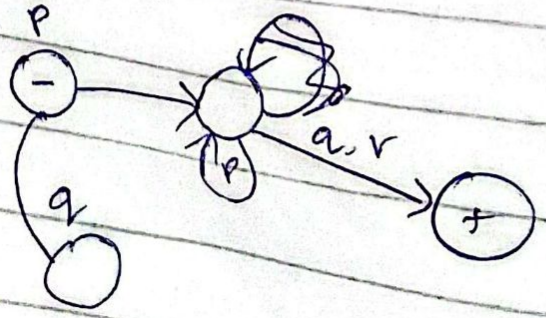
all the strings starting & ending with diff letters.



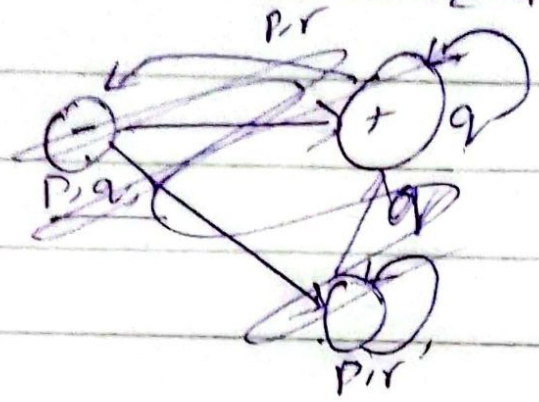
aaaab

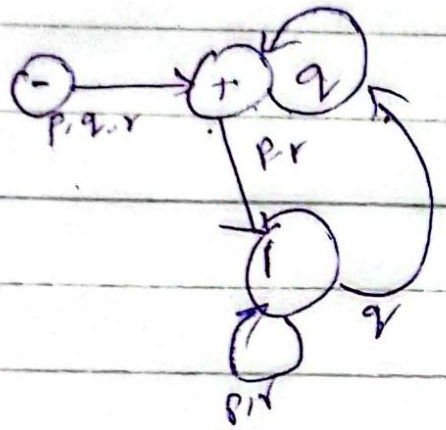


$\Sigma = \{p, q, r\}$

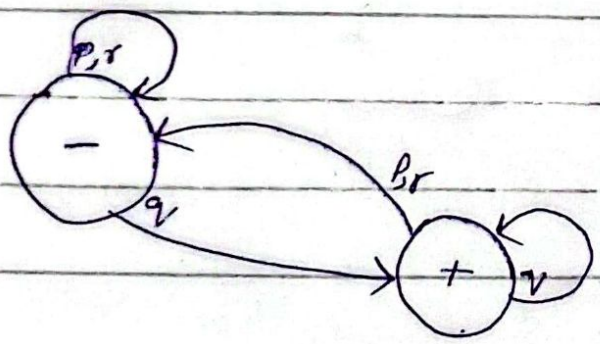


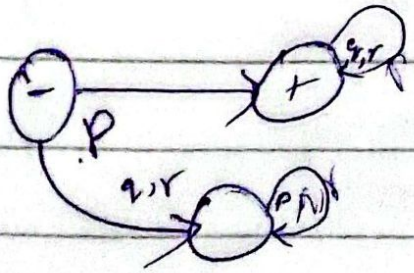
$\Sigma = \{p, q, r\}$ → end





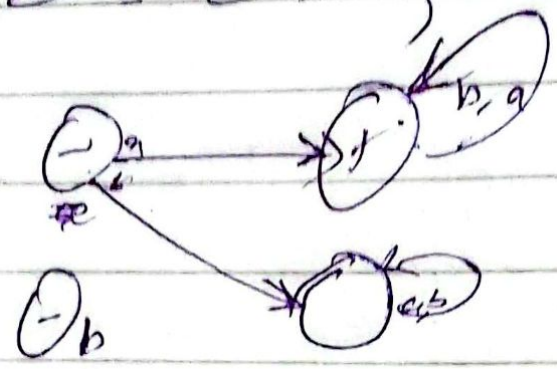
$\{p, q, r\}$ starting with p

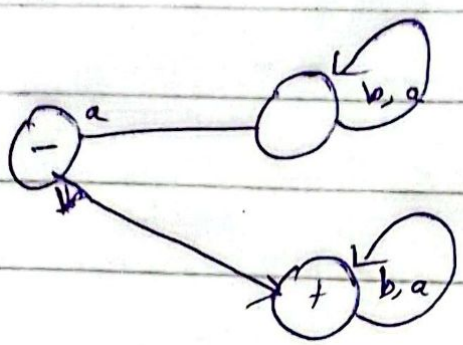




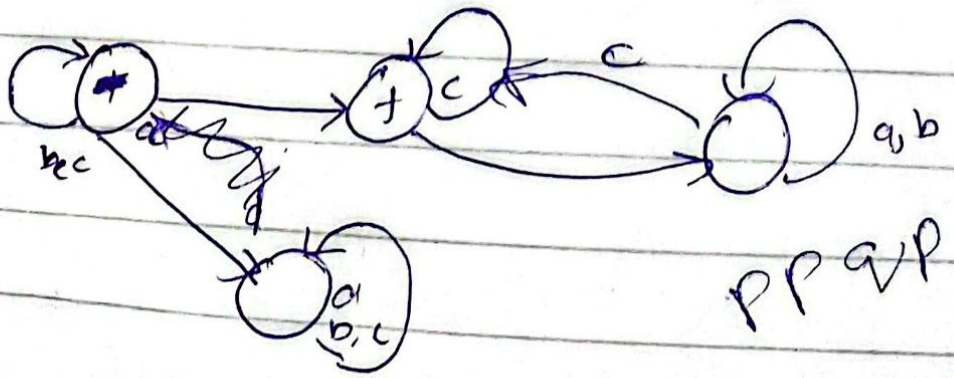
~~aab~~
bab

$\Sigma = \{a, b\}$

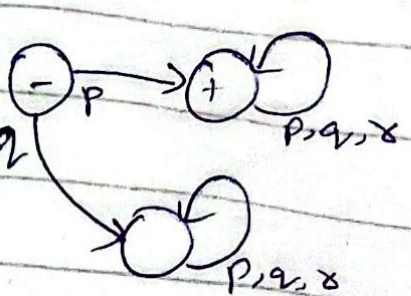


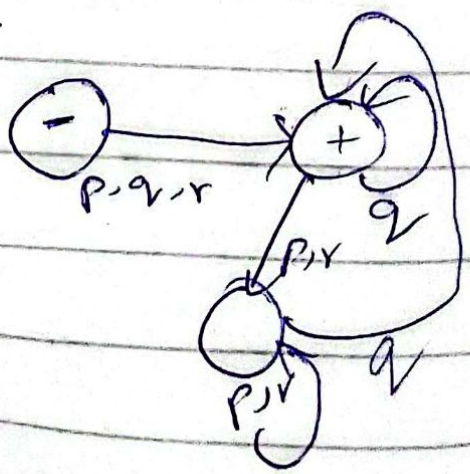


$\Sigma = \{a, b, c\}$



ppqvp





→ if a language contains such strings only; for such language for each string w

$$w = w^R$$

Rule #1 : $\wedge$, 'a', 'b' are valid strings of our language

Rule #2. If w is valid string of our language then 'awa, bwb'.

$\wedge$
a
b
aa
bb
aaa
bbb
aba
bab
aaaa
abba
bbbb
baab
aaaaa
abaaa
bbbbb
abbba
aabaa
bbabb
babab
baaab

Even Palindrome

Rule #1: $\wedge$ is valid string of our language.

Rule #2: If w is valid string of our lan———

Odd Palindrome

'a', 'b'———

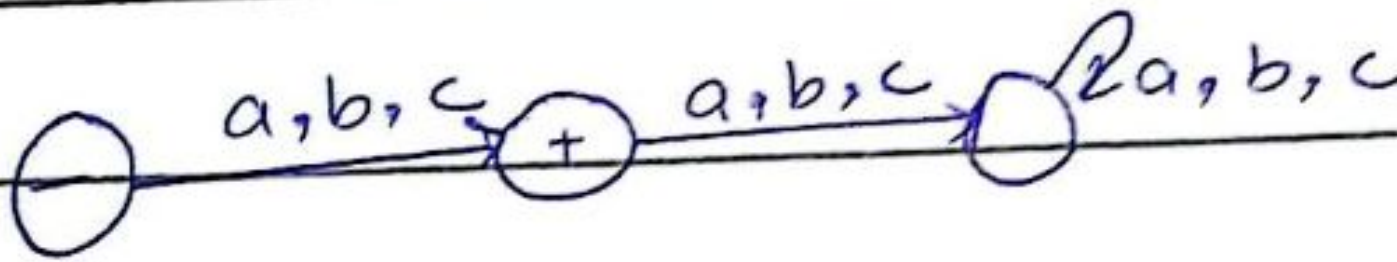
DATE:

DAY:

$\Sigma = \{a, b, c\}$

$L = \Sigma$

f_A :



$L = \phi$

string ka banai hoti



$L = \{\Lambda\}$



DAY: _____

DATE: _____